Javad Komijani

Research Scientist | Generative Models & Scientific ML

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Professional Summary

Research scientist working across generative modelling, statistical inference, and computational physics. I develop research ideas end to end, from mathematical formulation and algorithm design to PyTorch implementation, computational experiments, and quantitative validation. My work spans symmetry-aware diffusion and normalizing flows, reusable research software, and distributed training. I have also independently developed the theoretical framework, effective-field-theory model, and statistical analysis underlying precision quark-mass determinations using lattice-QCD data generated by the MILC Collaboration. My research combines mathematical depth with hands-on implementation and rigorous empirical testing.

Research impact: 27 peer-reviewed publications (full list), h-index 20, 3426 citations (Google Scholar, Oct 2026).

Selected Research Contributions

- **Theory, statistics & generative AI**: Developed a single-author framework for two-dimensional SU(N) lattice gauge sampling by reformulating plaquette-to-link consistency as a conditional sampling problem. Derived group-commutator densities for SU(2) and SU(3), combined analytical sampling with learned normalizing flows and importance-weight correction, and validated the method against Hybrid Monte Carlo (HMC). Tests cover 2×2 and 32×32 lattices, with reported per-plaquette acceptance rates above 99%. arXiv:2610.09147 | Reproducible notebooks
- **Independent research in computational physics**: Independently developed the theoretical framework, effective-field-theory model, and statistical analysis for precision determinations of five quark masses using lattice-QCD data generated by the MILC Collaboration. Addressed perturbative renormalon divergence and developed the minimal-renormalon-subtracted (MRS) mass framework to connect theoretical predictions with simulation results. This work resulted in a trilogy of papers and contributed to precision quark-mass determinations reported in FLAG/PDG reviews.
- **Efficient generative models**: Developed symmetry-aware normalizing flows and diffusion models, including Lie-group score matching, gauge-equivariant architectures, and analytically derived noise schedules. Designed an EFT-inspired Fourier-space layer with approximately 3.4K parameters and a 1,288-parameter SVD-based gauge-equivariant flow block that achieved effective sample size comparable to a 288-layer spectral-flow baseline on a 4^4 SU(3) lattice.

Research Experience

2021–2026 **Postdoctoral Researcher**, *ETH Zurich*
- Developed and evaluated generative-model algorithms end to end, from mathematical formulation and PyTorch implementation to numerical experiments and statistical validation.
- Built open-source research software and distributed training infrastructure for scientific ML, including differentiable numerical solvers and Lie-group-based sampling methods.
- Integrated normalizing-flow modules into the EU interTwin Digital Twin Engine.
- Co-supervised PhD, MSc, BSc, and intern projects in computational physics and scientific ML.

2015–2021 **Postdoctoral Researcher**, *TU Munich / Glasgow / Tehran*
- Monte Carlo simulations and Bayesian parameter estimation for precision lattice-QCD observables.
- Developed scalable lattice-QCD simulation pipelines on HPC systems in international collaborations.

Open-Source Scientific ML Software

lattice_ml **Composable PyTorch library for lattice field theory**

- Adjoint-method ODE integrators for memory-efficient backpropagation of derivatives through long trajectories, compatible with automatic differentiation, for scalar and Lie-group-valued objects; integrators that stay on the group manifold; symplectic solvers; Hutchinson trace estimation.
- Implicit score matching and diffusion on Lie groups with predictor–corrector sampling; flow matching and flow maps.
- HMC for U(1) and SU(N); AD-safe SVD and eigendecomposition; gauge-equivariant layers; prelink and holonomy tools; Lie-group coordinates and random-matrix sampling; distributed training.
- Used in arXiv:2605.06134, 2605.17326, and 2610.09147 (reproducible notebooks).

normflow **PyTorch package for normalizing flows in lattice field theory**

- Scalar and SU(N) gauge theories; self-learning from the action (reverse KL with path gradients) or data-based training; asymptotically exact sampling via a Metropolis step.
- EFT-inspired PSD and Fourier-space flows; coupling, rational-quadratic spline, and autoregressive layers.
- Spectral and SVD-based gauge-equivariant flows; distributed training.
- Used in arXiv:2301.01504, 2501.18288, 2605.06134, and 2610.09147.

Technical Skills

Programming	Python, C++, Cython, Bash; Linux, Git
ML/AI	PyTorch, Automatic Differentiation, Diffusion Models, Flow Matching, Normalizing Flows, Equivariant Deep Learning, Efficient Sampling (HMC, Predictor–Corrector), Distributed Training
Mathematics	Differential Equations, Linear Algebra, Lie Groups, Asymptotic Analysis, Optimization
Statistics	Monte Carlo Methods, Bayesian Inference, Importance Sampling, Statistical Modeling
Computing	CUDA, HPC and Supercomputing Environments, NumPy, SciPy
Languages	English (fluent), German (basic, A2)

Mentoring & Leadership

Mentoring **Co-supervised PhD, MSc, BSc, and intern researchers in physics & scientific ML:**

- PhD (2): gauge-equivariant diffusion models; Fourier acceleration for SU(N).
- MSc (6): equivariant U-Net; autoregressive, continuous-time, and spline-based flows; simulated tempering.
- BSc and intern: path-integral Monte Carlo; CNNs in arbitrary dimensions.

Organization **Co-organized workshops on physics & machine learning:**

- ZPW 2026 — Machine Learning Techniques for Particle Physics.
- QCD 2025 — Symposium on QCD Factorization.
- ML Lattice 2025 — Machine learning approaches in Lattice QCD.
- UT 2019 — Lattice QCD workshop; TUM 2017 — EFT and Lattice Field Theory Summer School.

Refereeing Referee for Physical Review D (13+ reviews since 2016).

Education

2015 **PhD in Physics**, *Washington University in St. Louis*
Thesis: *Topics in Lattice Gauge Theory and Theoretical Physics*
Advisors: Claude Bernard and Carl M. Bender

- Developed EFT-based frameworks and Bayesian inference methods for lattice QCD.
- Performed precision determinations of hadronic observables in lattice QCD.
- Developed a novel definition of eigenvalue problems for nonlinear systems arising in physics.

2009 **MSc in Electrical Engineering**, *University of Tehran*

- Focus: electromagnetic waves and fields, signal processing, statistical analysis.

2006 **BSc in Electrical Engineering**, *University of Tehran*